Department of Computer Science, Gujarat University

MCA123 Data Analytics

Assignment - Unit 1,2 & 3 (Practical)

1. Create a data frame of the below table.

Name	Age	Salary
Mahesh	23	10000
Kiran	25	12000
Suresh	28	15000
Sumit	26	14000
Ramesh	27	25000
Sawpnil	30	20000
Swati	29	26000

2. Get the maximum values of salary from the data frame.
3. Get the minimum values of the salary from the data frame.
4. Write the statement which will sort the marks in the Data Frame, in descending order.
5. Check the shape of data frame.
6. Add one row in the above data frame. (Name: Parita, Age: 32, Salary:25000)
7. Check the missing values in the data frame.
8. Display the data frame from Suresh to Parita by slicing.
9. Write a program to plot the ogive of random data from 1 to 20.
10. Write a program to plot the area chart of the data from 1 to 30.
11. Create the Data Frame of the below table.

Name	Maths	English	Science
Ramesh	78	67	56
Vedika	76	75	47
Harun	84	59	60
Prasad	67	72	54

12. Access the element in the 1^{st} row in the 3^{rd} column.
13. Access all the element in 3^{rd} column.
14. Access elements of 2^{nd} and 3^{rd} row from 1^{st} and 2^{nd} column.
15. Write a program to plot a pie chart of values=[3,4,7,9] and categories=['A', 'B', 'C', 'D'].
16. Create a Series from Dictionary. {'India':'New Delhi', 'UK': 'London', 'Japan':'Tokyo'}
17. Create a Series from 10 to 20 values using numpy and index should be of alphabet.
18. Write the code for outlier removal from scratch.

    [4, 7, 10, 14, 36, 16, 18, 20, 67, 22, 2]

19. Write the code of Finding Covariance from given data.

X= [2, 4, 6, 8, 10] , Y= [1, 3, 5, 7, 9]

20. Create a Data frame from Lists of Dictionary.

	a	b	c
0	10	20	Nan
1	5	10	20

21. Create a Data frame from Dictionary of Lists.

	State	Area	Temp.
0	Guj.	50123	30
1	Raj.	69536	35

22. Create a Data frame from Series.

	0
a	1
b	2
c	3
d	4
e	5

23. Create a Data frame from Dictionary of Series.

	Arnab	Ramit	Samridhi	Riya	Mallika
Maths	90	92	89	81	94
Science	91	81	91	71	95
Hindi	97	96	88	67	99

24. Do the following pre- processing techniques on the different 10 data sets.
   (a) Data Cleaning (Handling missing value for rows/columns, Handling Duplicates, Outliers detection and removal)
   (b) Handling Categorical data
   (c) Scaling of the data
   (d) Data Normalization
   (e) Identity insights which could be drawn from data and demonstrations of the same

Unit 2 and 3 (Book name- "Statistics for Business & Economics" by Anderson)

Examples: Exercise of each topic

Title	**Content**	**Book Chapter**
Data Visualization (Qualitative and Quantitative data)	Bar Charts, Pie chart, Scatter plots, line chart, area chart, ogive, stem and leaf, Dot Plot	**2**
Descriptive statistics	Measures of location, Measures of variability, Measures of association between two variables, Measures of distribution	**3**